

LLM-Based Crypto Market Signal Detection from X

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11. Mai 2026

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1 Introduction & Motivation

Cryptocurrencies have become a popular choice for investments. While many people assume these markets are secure, they are actually quite fragile and volatile. Public figures and influencers hold considerable power over market trends. A single post on social media can trigger drastic financial shifts. Consequently, the objective of this project is to apply Large Language Models to filter such posts and evaluate the exact impact they had on the market.

2 Project Goals

The project is structured around four main objectives:

- **Data Collection:** Extracting relevant posts utilizing the X API.
- **Filtering:** Developing an intelligent approach to query the API without exceeding resource limits or generating unnecessary expenses.
- **Detection:** Applying a Large Language Model to identify posts that could potentially trigger market movements.
- **Confirmation:** Evaluating historical price data to verify whether a specific post actually caused a measurable impact on the cryptocurrency market.

3 Architecture

The architecture of the system is divided into three main parts: data collection, processing through the LLM and the storage of the results.

3.1 Data Collection and Filtering

Due to API costs, data will not be loaded unfiltered. Instead we implement specific filters (e.g. for predefined accounts, certain hashtags or metadata like retweet/like counts). At first the project will focus on well known cryptocurrencies like Bitcoin and Ethereum. If we have enough time we will also include smaller coins. The filtered metadata and posts are then stored in a structured way within a relational database.

3.2 LLM Processing and Structured Output

For the semantic analysis we use a provided locally hosted LLM. To ensure the responses can be evaluated programmatically we will shape for machine output.